

Global Health Data Science Project 2026–2027

Suggested topic: Tap Water Quality

Alexandre Chanson & Veronika Peralta

1 Introduction

Access to safe drinking water is a fundamental pillar of global public health. However, agricultural practices, industrial activities, and urbanization pose ongoing threats to water quality. The intensive use of chemicals in agriculture, such as pesticides, fertilizers, and herbicides, frequently results in soil leaching and the contamination of groundwater aquifers.

As these water sources are drawn upon for public distribution, robust and systematic monitoring becomes critical to prevent long-term public health crises. This project aims to leverage data science techniques to analyze tap water quality, map its relationship with agricultural practices, and evaluate the environmental health risks across different regions. By cross-referencing health inspections with agrochemical data, you will explore the spatial and temporal correlations between human activity and the safety of our drinking water.

2 Provided data

2.1 Tap water analysis results

<https://www.data.gouv.fr/datasets/resultats-du-controle-sanitaire-de-leau-distribuee-commune-par-commune>

This dataset is extracted from the French Ministry of Health’s national database (SISE-Eaux). It contains the official results of sanitary controls on distributed tap water, broken down commune by commune across France. The data encompasses over 300,000 annual water samples and millions of laboratory analyses. It tracks microbiological, physicochemical, and radiological parameters to ensure local water supplies comply with European safety directives (Directive 98/83/CE).

2.2 Use of chemicals in agriculture

https://solagro.org/medias/domaines-intervention/agroecologie/Guide_me__thodologique_-_Carte_sur_l_usage_des_pesticides.pdf

This methodological guide by Solagro details the approach used to map and quantify pesticide usage in French agriculture, which they did for 2020, 2021, and 2022. Understanding the spatial distribution and intensity of these agricultural practices is crucial for correlating regional farming activities with the presence of chemical residues and metabolites found in local water distribution networks.

Data package The provided data is available as a single ZIP archive on the CELENE course platform.

3 Suggested data

In addition to the provided data sources, you may find other sources that could broaden your analysis or be used to make useful comparisons. Here are a few examples of sources that could prove useful:

Older data Older data is available but only tracks some pest control chemicals used in agriculture. Extracted from the Ministry of Ecological Transition, it covers the contamination of groundwater by tracking nearly 600 active substances (herbicides, insecticides, fungicides) and their degradation metabolites across approximately 2,200 monitoring stations (historically focusing on the 2007–2012 vintages). It can be found at: <https://www.data.gouv.fr/datasets/pesticides-dans-les-eaux-souterraines>

England The English government has a very similar approach to French authorities with systematic testing of tap water and public reporting of analysis results. Through the Department for Environment, Food & Rural Affairs (DEFRA) Water Quality Explorer, you can access sampling points and observation data for interesting cross-border comparisons. The data can be found at: <https://environment.data.gov.uk/water-quality>

Chemicals sales data Older data on sales (not actual use) of agricultural chemicals in agriculture per commune <https://www.data.gouv.fr/datasets/achats-de-pesticides-par-code-postal-1>